# UGED1111 Logic

# Natural Deduction I

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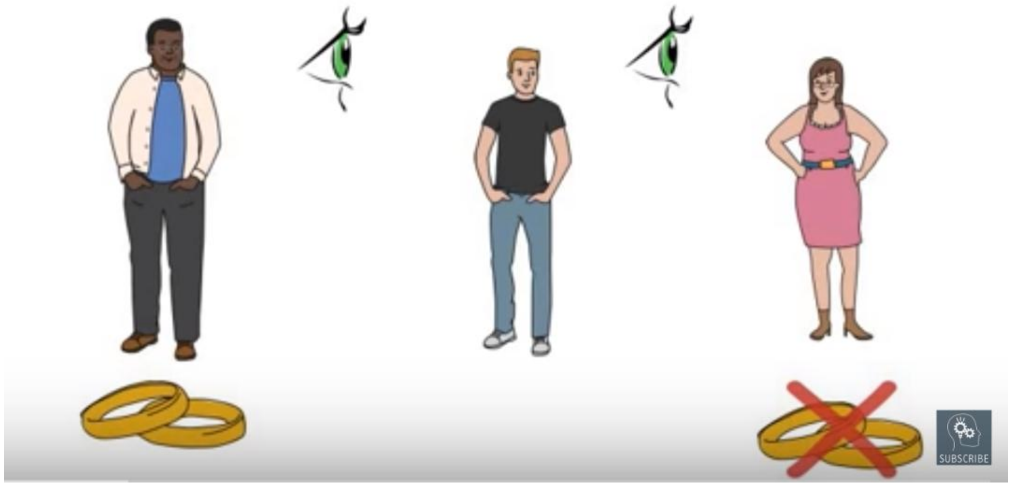
# Scenario

Is a married person looking at an unmarried person?

A: Yes

B: No

C: Not enough information



# Scenario

1. 當小蘭在廚房或小東不在廚房，則廚師會在廚房。
2. 只要小東在廚房，則小北也在廚房。 。
3. 並不是小北或小七在廚房。

4. 廚師在廚房。

1. $(P \vee \sim R) \supset S$
2. $R \supset Q$
3. $\sim (Q \vee M) // S$

P:小蘭在廚房。
Q:小北在廚房。
R:小東在廚房。
S:廚師在廚房。
M: 小七在廚房

# Checking validity

| (P | v | ~ R) ⊃ S / R ⊃ Q / ~ (Q v M) // S |
|---|---|---|
| | | |

Valid argument.

But is this all that we wants to know???

We also want to know how he deduces the conclusion from the premises!

# Natural Deduction

- Natural deduction is a method of deriving the conclusion of a valid argument from the premises through a series of discrete steps.

- This method aims to demonstrate the reasoning process of an argument (how we prove the conclusion from the premises).

- However, it does not work for invalid arguments.

# 18 Rules of Inference

**There are 18 rules of inference.**

- **8 Rules of Implication**

**+**

- **10 Rules of Replacement**

*****All 18 rules will be given in the exam. So you don’t need to bother memorizing them.**

# Rules of implication I

1. Modus Ponens (MP)

**p⊃q**
**p**
**q**

2. Modus Tollens (MT)

**p⊃q**
**~q**
**~p**

3. Hypothetical Syllogism (HS)

**p⊃q**
**q⊃s**
**p⊃s**

4. Disjunctive Syllogism (DS)

**p v q**
**~p**
**q**

# Rules of implication II

5. Constructive Dilemma (CD)

$(p \supset q) \cdot (r \supset s)$

$p \vee r$

---

$q \vee s$

6. Simplification (Simp)

$p \cdot q$

---

$p$

7. Conjunction (Conj)

$p$

$q$

---

$p \cdot q$

8. Addition (Add)

$p$

---

$p \vee q$

# 1. Modus Ponens (MP)

Given a conditional statement , we can deduce its consequent by asserting its antecedent.

$p \supset q$
$p$
$\overline{\phantom{p \supset q}}$
$q$

1. If you get GPA 4，you will get the scholarship。
2. You get GPA 4 。

---

3. You will get the scholarship 。

# 2. Modus Tollens (MT)

Given a conditional statement and the negation of its consequent, then we can deduce the negation of its antecedent.

$$\begin{array}{l} p \supset q \\ \sim q \\ \hline \sim p \end{array}$$

1. If the temperature is higher that 30 degree Celsius ， then the school will turn on the air-conditioner。
2. The school did not turn on the air-conditioner today。

3. The temperature is not higher than 30 degree Celsius today.

# 3. Hypothetical Syllogism (HS)

Given two conditional statements with a common formula as transition then we can deduce a new conditional without that common formula.

$$\frac{\begin{array}{l} p \supset q \\ q \supset s \end{array}}{p \supset s}$$

1. 如果我收到利是，則母親幫我保存利是。
2. 如果母親幫我保存利是，則利是變成母親存款。

---

3.如果我收到利是，則利是變成母親存款。

# 4. Disjunctive Syllogism (DS)

Given a disjunctive statement and the negation of its left-hand disjunct, we can deduce the right-handed disjunct.

| | |
|---|---|
| $p \vee q$ | 1. 小明是帥哥或者有才華的。 |
| $\sim p$ | 2. 小明不是帥哥。 |
| $q$ | 3. 小明是有才華的。 |

# Discussion 1a

Construct a formal proof of validity for the following arguments by Natural Deduction.

1. $P \supset Q$
2. $Q \supset S$
3. $\sim S \quad // \sim P$

# Discussion 1b

Construct a formal proof of validity for the following arguments by Natural Deduction.

1. $P \vee S$
2. $S \supset Q$
3. $\sim P \quad //Q$

# 5. Constructive Dilemma (CD)

Given a conjunction of two conditional statements. If either of the antecedents is true, then either of the consequents is true.

$$\frac{\begin{array}{l}(p \supset q) \cdot (r \supset s) \\ p \vee r\end{array}}{q \vee s}$$

1. 如果下雨則賣水鞋，並且如果天晴則賣傘。
2. 下雨或天晴。

---

3. 賣水鞋或賣傘。

# 6. Simplification (Simp)

Given a conjunction of two statement, then we can deduce the left-sided conjunct.

$$\frac{p \cdot q}{p}$$

1. 殺老師反應很快，而且他很聰明。

---

2.殺老師反應很快。

# 7. Conjunction (Conj)

Given that two propositions are true separately, then their conjunction are also true.

| | |
|---|---|
| p | 1. I have a pen. |
| q | 2. I have an apple. |
| ――――― | ――――――――――――― |
| p · q | 3. I have a pen and a apple. |

# 8. Addition (Add)

Given a proposition is true, its disjunction with any proposition is also true.

$$\frac{p}{p \vee q}$$

1. Karina很漂亮。

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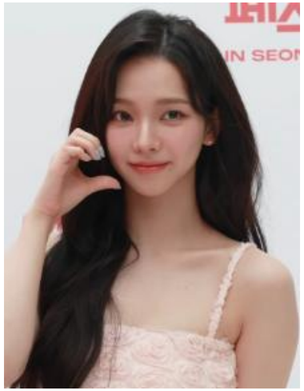
2. Karina很漂亮，或者Karina很可愛。



# Discussion 2

Construct a formal proof of validity for the following arguments by Natural Deduction.

1. $P \cdot M$
2. $P \supset Q$
3. $R \supset S$ // $Q \vee S$

# Exercise 1

1. $\sim C \supset (A \supset C)$
2. $\sim C \qquad\qquad\qquad / / \sim A$

# Exercise 2

1. $\sim S \supset D$
2. $\sim S \vee (\sim D \supset K)$
3. $\sim D$ $\qquad\qquad\qquad$ // $K$

# Invalid inference

**Rules of implication cannot be applied on a part of compound statements.**

1. $P \supset Q$
2. $(P \vee R) \supset Q$ 1, Add

1. $(P \cdot R) \supset Q$
2. $P \supset Q$ 1, Simp

1. $P \supset Q$
2. $(P \supset Q) \vee R$ 1, Add

1. $(P \supset Q) \cdot R$
2. $P \supset Q$ 1, Simp

# Improper use of rules

**Even though the following inferences are valid, they are improper.**

1. $P$
2. $Q \vee P$     1, Add

1. $P \cdot R$
2. $R$     1, Simp

1. $P$
2. $P \vee Q$     1, Add

1. $P \cdot R$
2. $P$     1, Simp

# Rules of replacement I

9. DeMorgan's Rule (DM)

$\sim(p \cdot q) :: \sim p \vee \sim q$

$\sim(p \vee q) :: \sim p \cdot \sim q$

10. Commutativity (Com)

$p \vee q :: q \vee p$

$p \cdot q :: q \cdot p$

11. Association (Assoc)

$p \vee (q \vee r) :: (p \vee q) \vee r$

$p \cdot (q \cdot s) :: (p \cdot q) \cdot s$

12. Distribution (Dist)

$p \cdot (q \vee r) :: (p \cdot q) \vee (p \cdot r)$

$p \vee (q \cdot s) :: (p \vee q) \cdot (p \vee s)$

13. Double Negation (DN)

$p :: \sim\sim p$

# 9. DeMorgan's Rule (DM)

**"Not both p and q" is logically equivalent to "Not p or not q" ;**

and

**"Not either p or q" is logically equivalent to "Not p and not q."**

$\sim(p \cdot q) :: \sim p \vee \sim q$

並不是大嗺鬼和「任何人」都有頭髮。
大嗺鬼沒有頭髮或「任何人」沒有頭髮。

$\sim(p \vee q) :: \sim p \cdot \sim q$

並不是大嗺鬼或「任何人」有穿衣服。
大嗺鬼沒有穿衣服而且「任何人」沒有穿衣服。


# 10. Commutativity (Com):

The truth value of a conjunction or disjunction is unaffected by the order in which the components are listed.

$\mathbf{p \lor q \;::\; q \lor p}$

今天是晴天，或者今天是陰天。
今天是陰天，或者今天是晴天。

$\mathbf{p \cdot q \;::\; q \cdot p}$

熊本熊和鬆弛熊來中大。
鬆弛熊和熊本熊來中大。

# 11. Association (Assoc)

When the same operator is used through out a conjunctive or disjunctive statement, the placement of brackets will not alter the truth value of the whole statement.

| | |
|---|---|
| **$p \vee (q \vee r) :: (p \vee q) \vee r$** | 今天是晴天，或者今天是陰天或雨天。<br>今天是晴天或陰天，或者今天是雨天。 |
| **$p \cdot (q \cdot s) :: (p \cdot q) \cdot s$** | 北極熊來中大，而且黑熊和啡熊來中大。<br>北極熊和黑熊來中大，而且啡熊來中大。 |

# 12. Distribution (Dist):

When a proposition is conjoined to a disjunctive statement in bracket or disjoined to a conjunctive statement in bracket, the rule allows us to put that proposition together with each of the components inside the bracket, and also to go in the reverse direction.

$$p \cdot (q \vee r) :: (p \cdot q) \vee (p \cdot r)$$

北極熊來中大，並且黑熊或啡熊來中大。
北極熊和黑熊來中大，或者北極熊和啡熊來中大。

$$p \vee (q \cdot s) :: (p \vee q) \cdot (p \vee s)$$

**Similar technique you have used before**

**$2 \times (4+3) = (2 \times 4)+(2 \times 3)$**

**$m \times (n+y) = mn + my$**

# 13. Double Negation (DN)

If pairs of tildes immediately adjacent to one another, deleting or introducing them cannot change the truth value of the whole statement.

**p :: ~~p**

我沒有不上課。
我有上課。

**負負得正?!**

# Discussion 3

1. $\sim(\sim P \vee Q)$
2. $P \supset S \qquad //S$

# Discussion 4

1. $D \bullet (E \vee F)$
2. $\sim D \vee \sim F \quad // \; D \bullet E$

# Exercise 3

1. $(K \bullet H) \vee (K \bullet L)$
2. $\sim L$ // $H$

# Exercise 4

1. $S \vee (P \cdot Q)$
2. $(S \vee Q) \supset \sim R$
3. $(R \vee P) \vee Q$ $\qquad$ $// P \vee Q$

# 14. Transposition (Trans):

The antecedent and consequent of a conditional statement may switch places if and only if tildes are inserted before both or tildes are removed from both.

$$(p \supset q) :: (\sim q \supset \sim p)$$

如果昨天我睡得夠，則我今天上課很清醒。
如果我今天上課不是很清醒，則我昨天睡不夠。

# 15. Material Implication (Impl)

**Material implication** indicates the truth condition of a conditional statement, which means that a conditional statement is true if and only if the antecedent is false or the consequent is true

$$(p \supset q) :: (\sim p \vee q)$$

如果我口渴，則我喝水。
我不是口渴，或者我喝水。

# 16. Exportation (Exp)

"If we have both p and q, then we have r"
is logically equivalent to
"If we have p, then if we have q, then we have r."

$(p \cdot q) \supset r :: p \supset (q \supset r)$

如果你有車和有樓時，我們結婚。

如果你有車，則當你有樓時，我們結婚， 。

# Recall

**Rule of replacement**
**Exportation**
**$(p \cdot q) \supset r :: p \supset (q \supset r)$**

如果你有車和有樓時，則我們結婚。

如果你有車，則當你有樓時，我們結婚， 。

**Consider the following case of using rule of implication**

1. **$(P \cdot Q) \supset R$**
2. **$P \supset R$** **1, Simp**

1. 如果你有車和有樓時，我們結婚。
2. 如果你有車，我們結婚。 1, Simp

# 17. Material Equivalent (Equiv):

(i) The first is the same as the definition of material equivalence given in the previous class.

(ii) The second formulation indicates the truth condition of "**$p \equiv q$** ", which means "**$p \equiv q$** " *is true iff **either p and q are both true*** **or** ***p and q are both false".***

$$(p \equiv q) :: (p \supset q) \cdot (q \supset p)$$

$$(p \equiv q) :: (p \cdot q) \vee (\sim p \cdot \sim q)$$

# 17. Material Equivalent (Equiv):

我獲得獎學金當且僅當我GPA 4.

(1)我獲得獎學金當我GPA 4，而且當我獲得獎學金則我GPA 4

(2)我獲得獎學金並且我GPA 4，或者我沒有獲得獎學金並且我不是GPA 4.

$$(p \equiv q) :: (p \supset q) \cdot (q \supset p)$$

$$(p \equiv q) :: (p \cdot q) \vee (\sim p \cdot \sim q)$$

# 18. Tautology (Taut):

To eliminate redundancy in disjunctions and conjunctions.

| | |
|---|---|
| **p :: (p ∨ p)** | 我好想睡或者我好想睡。 |
| **p :: (p · p)** | 我好肚餓而且我好肚餓。 |

# Rules of replacement II

14. Transposition (Trans)

$$(p \supset q) :: (\sim q \supset \sim p)$$

15. Material Implication (Impl)

$$(p \supset q) :: (\sim p \ v\ q)$$

**16.** Exportation **(Exp)**

$$(p \cdot q) \supset r :: p \supset (q \supset r)$$

17. Material Equivalent (Equiv)

$$(p \equiv q) :: (p \supset q) \cdot (q \supset p)$$

$$(p \equiv q) :: (p \cdot q)\ v\ (\sim p \cdot \sim q)$$

18. Tautology (Taut)

$$p :: (p \vee p)$$

$$p :: (p \cdot p)$$

# Discussion 5

1. $S \supset (\sim S \vee P) \quad // \; S \supset P$

# Discussion 6

1. $P \equiv Q$
2. $Q \supset (R \cdot S)$
3. $\sim S$ $\qquad$ // $\sim P$

# Discussion 7

1. $P \supset Q$
2. $S \supset M$
3. $\sim Q$ // $\sim P \vee M$

# Reference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